

Assignment - 9

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Problem - 1 Training an autoencoder as a 2D feature generator and displaying CIFAR10 dataset's features
Below given the output for the 2D feature generator for the Auto-Encoder

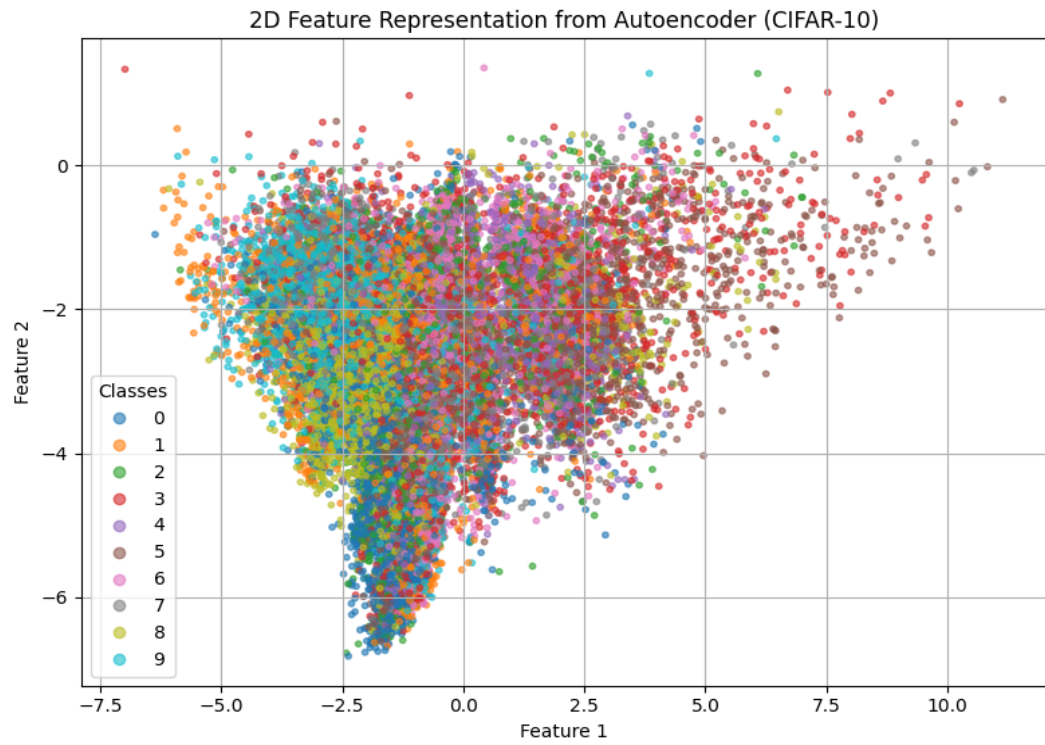


Figure 1: Output of Auto-Encoder

Problem - 2 comparing autoencoder generated features with features extracted by a pre-trained CNN and reduced by dimension reduction techniques like PCA, t-SNE.
Below given the output after comparing different models result

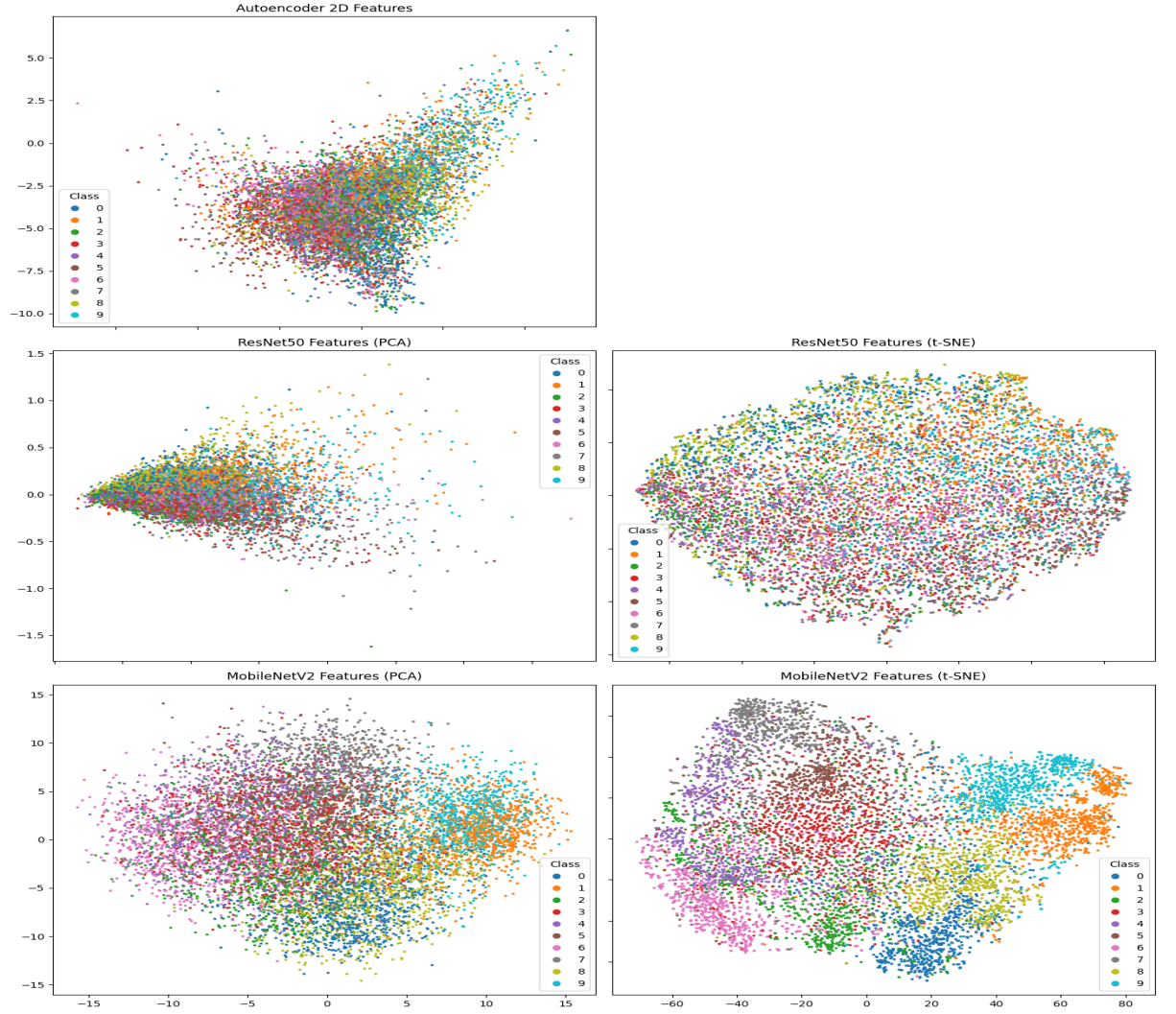


Figure 2: Output of different models for PCA and t-SNE

Problem - 3 Training a denoising autoencoder for CIFAR10 dataset.
Below given the output for the Denoising Auto-Encoder.

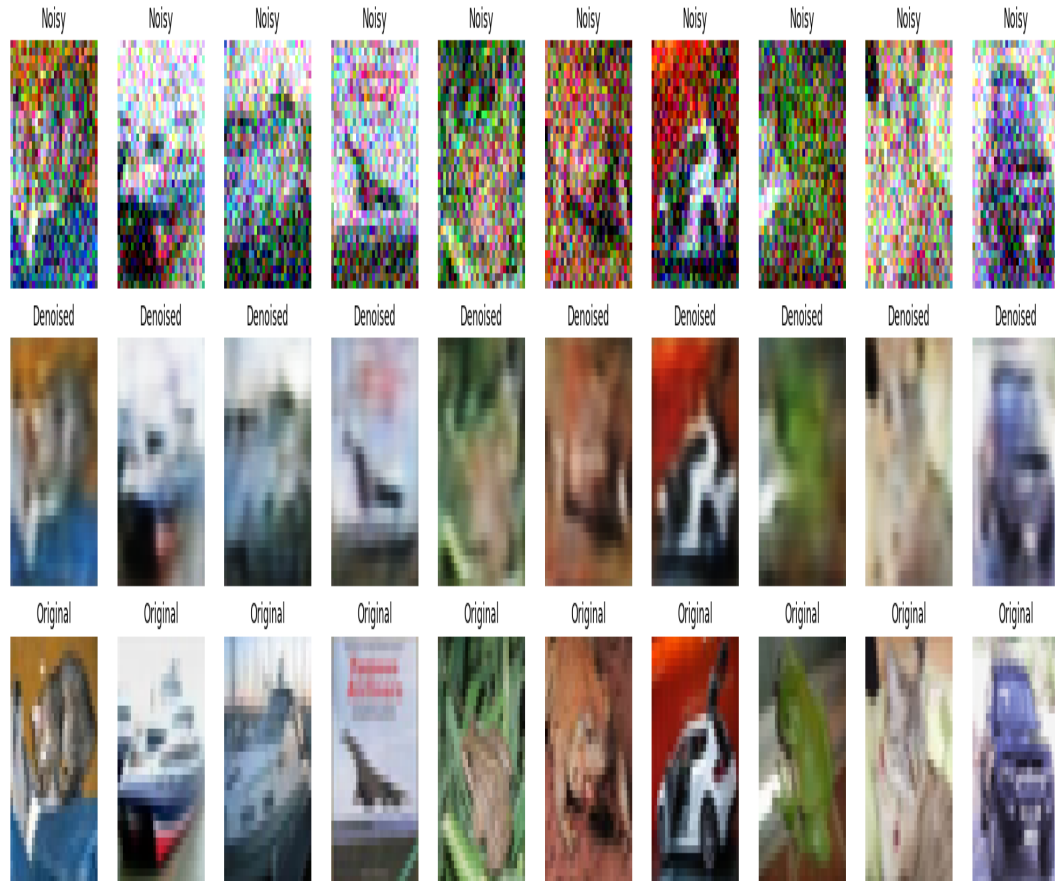


Figure 3: Output of Denoising Auto-Encoder

Problem - 4 Training a CNN based CIFAR-10 classifier without any single-image data augmentation techniques.
Below given the output for the CNN based augmentation result.

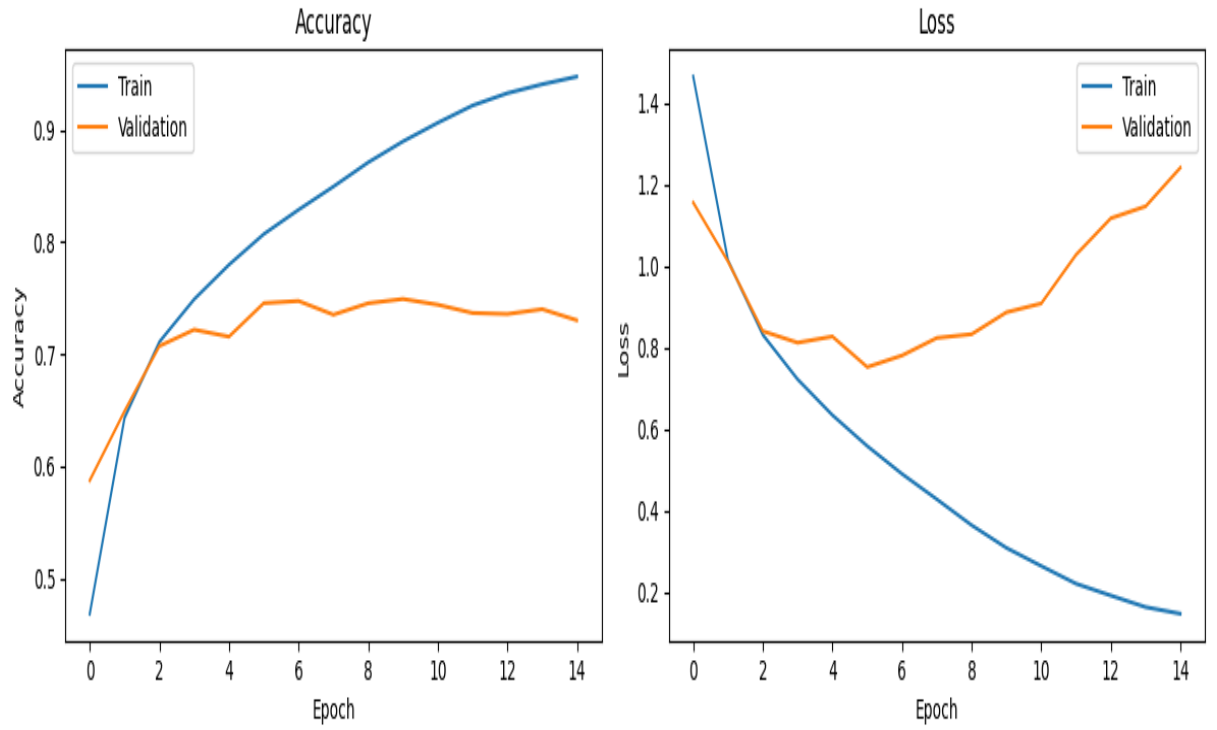


Figure 4: Output of Single-image augmentation

Problem - 5 Training a CNN based CIFAR-10 classifier with a single/multiple single-image data augmentation techniques
Below given the output for the CNN-base single/multiple classifier augmentation result.

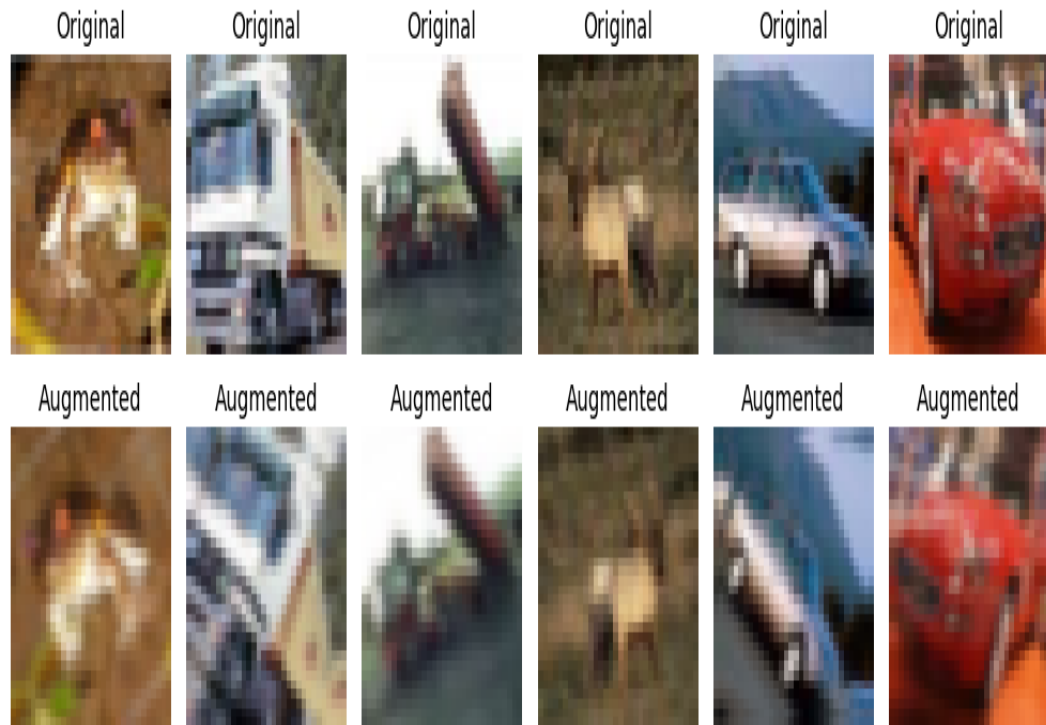


Figure 5: Output of Multiple augmentation image

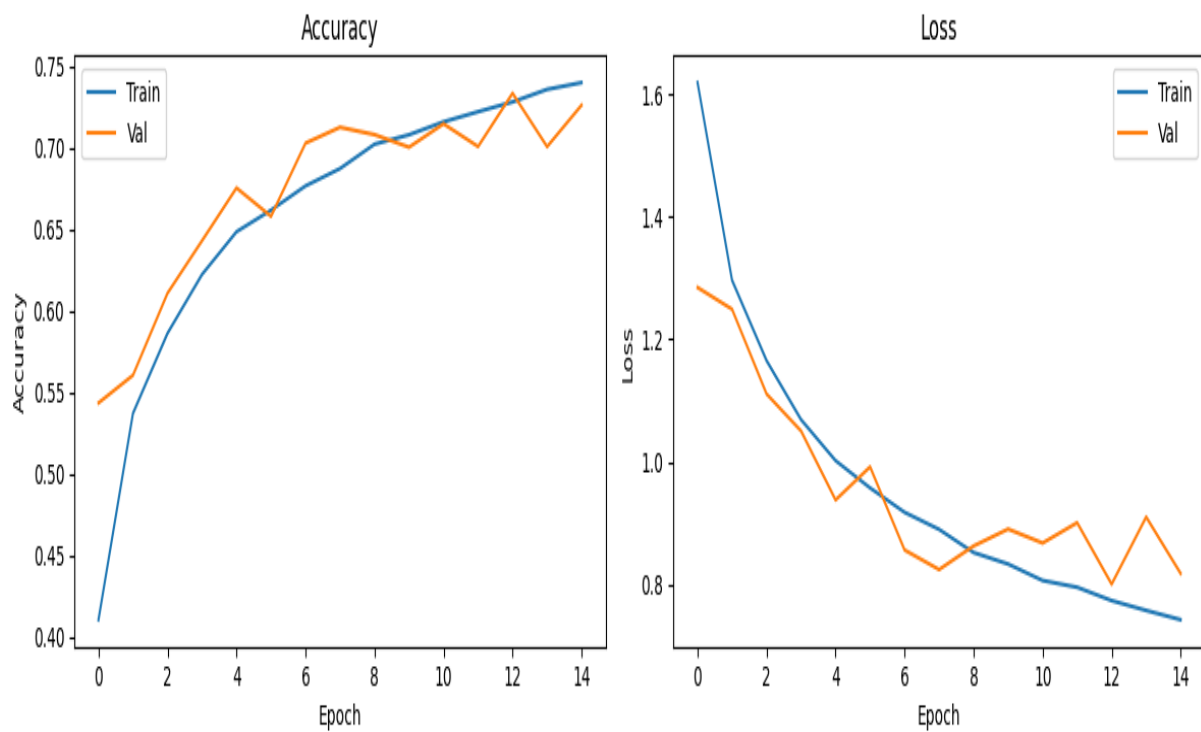


Figure 6: Output of Multiple augmentation images result

Assignment Github Link : Assignment